96%.

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: **CSA16 COURSE CODE: Data Warehousing and Data Mining**

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:20**

Annexure A
SHORT ANSWER TEST

Criteria	**Conceptual Understanding(3)**	**Accuracy of Answer(2.5)**	**Application of Knowledge(2)**	**Completeness of Response(1.5)**	**Clarity & Presentation(1)**	*Total(10)*
Weightage	*30%*	25%	20%	15%	10%	100%
Q1	~~30~~ 3	~~20~~ 2	~~20~~ 2	~~15~~ 1.5	~~10~~ 1	9.5
Q2	3	2.5	2	1	1	9.5
Q3	3	2	2	1.5	1	9.5
Q4	3	2.5	2	1	1	9.5

95%

Code of Conduct:

I PARVEEN [192462005] *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Parveen
Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Annexure A

Short Answer Test

1.Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.

(b) Draw a schema diagram for the above data warehouse using one of the schema.

(c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).

(5 Marks)

2.Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [student,course, semester,instructor], what specific OLAP operations

(e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses

for each Big University student.

(c) If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain (including the base and apex cuboids)?

3.Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space. (5 Marks)

4. A data cube, C, has n dimensions, and each dimension has exactly p distinct values in the base cuboid. (5 Marks)

Assume that there are no concept hierarchies associated with the dimensions.

(a) What is the maximum number of cells possible in the base cuboid?

(b) What is the minimum number of cells possible in the base cuboid?

(c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the

data cube, C?

(d) What is the minimum number of cells possible in the data cube, C?

Question 1 : Design a Data Warehouse for Doctor-Patient Visits

95

Introduction

- A Data Warehouse is a centralized repository that stores large volumes of integrated, historical and subject-oriented data collected from different operational databases.

(a) Three Popular Data Warehouse Schemas

Schema	Description	Advantages
Star Schema	One central fact table connected directly to dimension tables.	Simple, easy to understant, fast query processing.
Snowflake Schema	Dimension tables are normalized into multiple related tables.	Reduces redundancy and improves data integrity.
Fact Constellation Schema	Multiple fact tables are used to share common dimension tables.	Suitable for complex organization with the multiple business processes.

Comparison of Schemas

Feature	Star Schema	Snowflake Schema	Galaxy Schema
Structure	Simple	Normalized	Complex
Query Speed	High	Meduim	Medium
Redundancy	More	Less	Moderate

(b) Star Schema Design

Time Dimension	
Attributes	
1.	Day
2.	Month
3.	year

Doctor Dimension	
Attributes	
1.	Doctor ID
2.	Name
3.	Hospital
4.	Specialization

Fact_Visit	
Measures & Foreign Keys	
1.	Time_ID (FK)
2.	Doctor_ID (FK)
3.	Patient_ID (FK)
4.	Count
5.	Charge

Patient Dimension	
Attributes	
1.	Patient ID
2.	Name
3.	Age
4.	Gender

(c) SQL Query

SELECT day, month, year, doctor, hospital, patient, count, carge
FROM fee;

Advantages of Star Schema

- Easy to understand
- Fast query performance
- Simple database design.

Conclusion

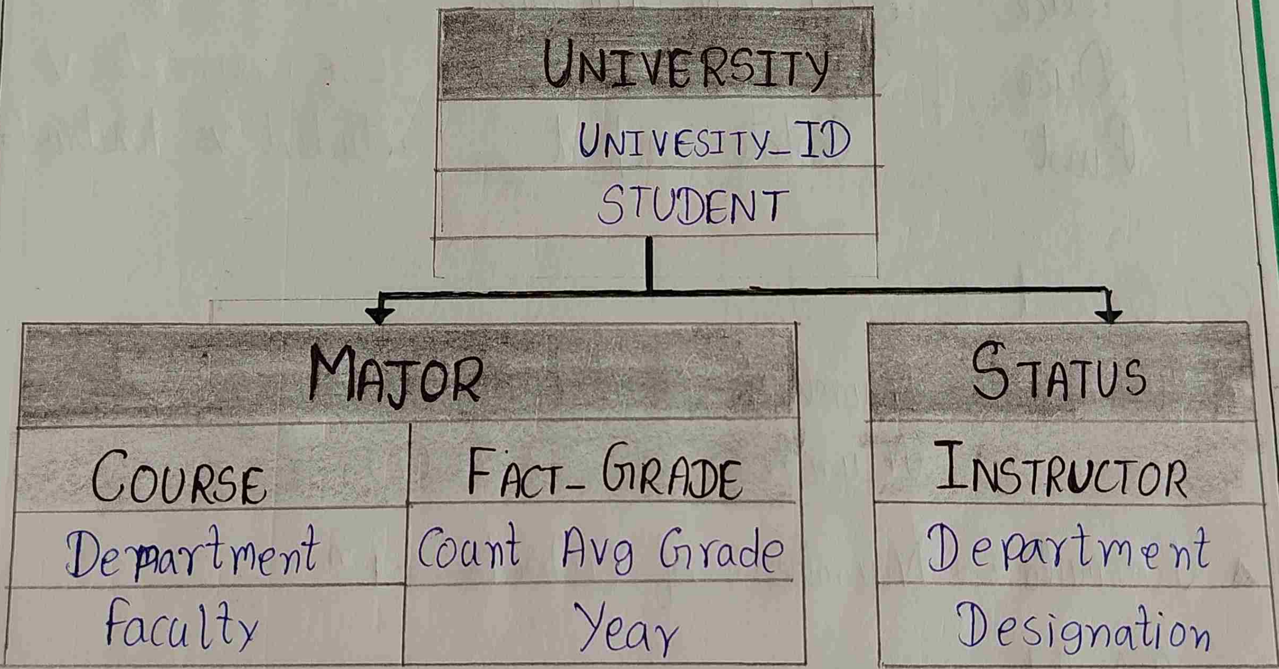
Using measures Count and generate meaningful reports and make informed descions.

Question 2: Design a Snowflake Schema for Big University

Introduction

- A univerity Data Warehouse stores academic information in an organized form so that management can be analyze student performance, course, instructor.

(a) Snowflake Schema



Fact Table

Attribute	Description
Student_ID	Student Key
Course_ID	Course Key
Semester_ID	Semenster Key
Instructor_ID	Instructor Key
Count	No. of students

(b) OLAP Operations

- To display the average grade of CS courses for each student, the following OLAP operations are performed

Operation	Purpose	Example
Roll-up	Summarize data	Semester → year
Drill-down	View detailed data	year → Semester
Slice	Select one dimension	CS course
Dice	Filter multiple values	CSE + semester V
Pivot	Rotate report	Student vs Instructor

(c) Cuboid Calculation

▲ Given : * Number of Dimensions $(n) = 4$
* Number of Levels $(L) = 5$

▲ Formula : * Number of Cuboids $= L^n$

▲ Calculation :

$$L^n = 5^4$$
$$= 5 \times 5 \times 5 \times 5$$
$$= 25 \times 25$$
$$= 625$$

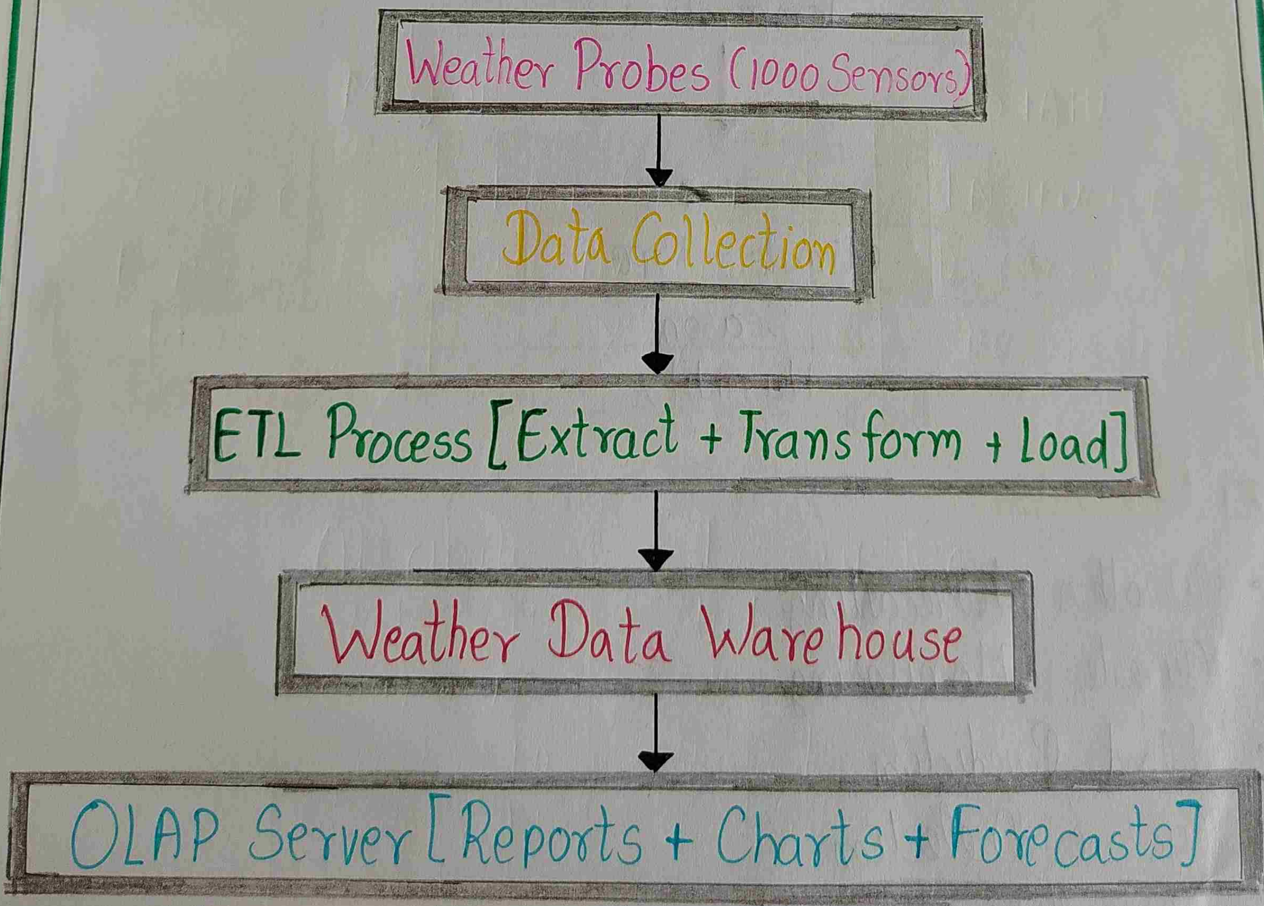
- The cube contains 625 cuboids, including base cuboid and apex cuboid.

Question 3 : Design a Data Warehouse for a Regional Weather Bureau.

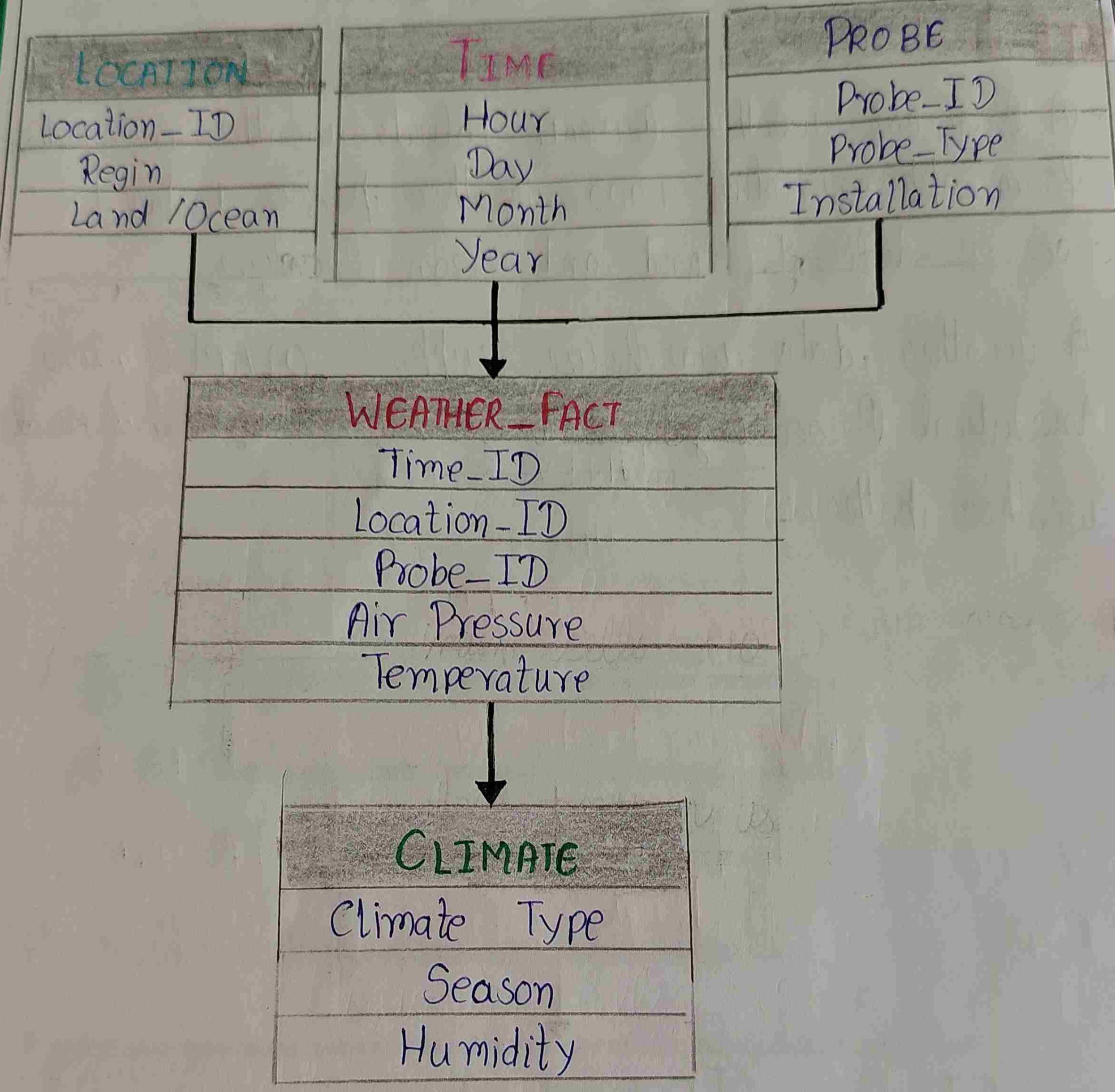
Introduction:

- A Regional weather bureau collects weather information from approximately 1,000 weather probes installed across different land and ocean locations.
- A weather data warehouse supports OLAP [Online Analytical Processing], allowing to analyze historical weather patterns.

Data Warehouse Architecture



Star Schema Diagram



Applications

- Weather Forecasting
- Climate Monitoring
- Flood Prediction
- Cyclone Detection
- Agriculture Planning.

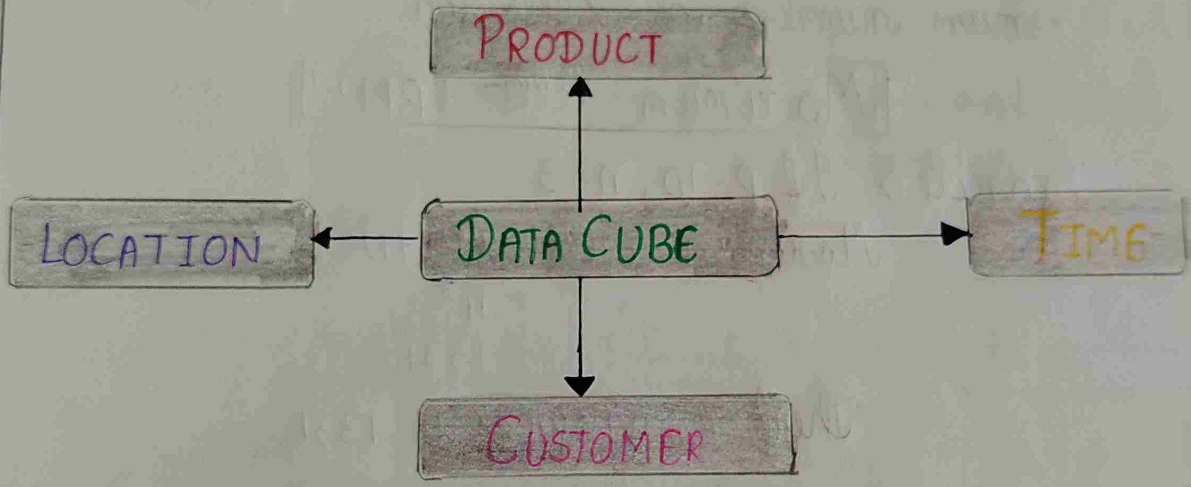
Question 4: Data Cube and Cell Calculations

Introduction

- A Data Cube is a multi-dimensional representation of Data used in a Data Warehouse. It enables users to analyze different information.

Basic Concept of Data Cube

- A data cube organizes data in multiple dimensions



(a) Maximum number of Cells in Base Cuboid

- Formula: $\boxed{\text{Maximum Cell} = p^n}$

Exampl ⇒ Let $p = 10$, $n = 3$

$$\text{Maximum Cells} = 10^3$$

$$= 1000$$

Hence answer is p^n //

(b) Minimum number of Cells in Base Cuboid

• Formula : $\boxed{\text{Maximum Cells} = 1}$

Example ⇒ Product = Laptop

Location = Chennai

Time = January

Only 1 cell information

Therefore Answere is 1,,

(c) Maximum number of cells in Entire data cube

• Formula: $\boxed{\text{Maximum Cells} = (p+1)^n}$

Example ⇒ If $p = 10$, $n = 3$

$$\text{Maximum cells} = (10+1)^3$$
$$= 11^3$$
$$= 1331$$

Therfore answer is 1331,,

(d) Minimum number of Cells in Entire Data Cube

• Formula: $\boxed{\text{Minimum Cells} = 2^n}$

Example ⇒ If $n = 3$

$$\text{Minimum Cells} = 2^3$$
$$= 8$$

Therefore answer is 8,,